

Multiple Choice (10 marks)

1. Pat is learning how energy flows through an ecosystem. Pat understands that animals get energy from plants. What do plants do that allow animals to get energy from them?
 - (a) Plants keep water in their roots.
 - (b) Plants release nitrogen into the soil.
 - (c) Plants provide shelter from the weather.
 - (d) Plants make sugar in the presence of light.
2. Which of these is an example of an adaptation for defense from predators?
 - (a) giraffe's long neck
 - (b) baby deer's spots
 - (c) otter's oily fur
 - (d) fish's gills
3. Which planet in the solar system has the longest year?
 - (a) The planet closest to the Sun.
 - (b) The planet with the longest day.
 - (c) The planet with the most moons.
 - (d) The planet farthest from the Sun.
4. A student poured water into a plastic tray. The student then put the tray into the freezer. What property of the water changed as the water froze?
 - (a) The water became a gas.
 - (b) The mass of water increased.
 - (c) The water took a definite shape.
 - (d) The flavor of the water changed.
5. Rossby waves are slow-moving, long ocean waves. What feature of a Rossby wave would change if the wavelength was doubled?
 - (a) The amplitude would double.
 - (b) The frequency would double.
 - (c) The amplitude would decrease by half.
 - (d) The frequency would decrease by half.

True / False (10 marks)

Answer True or False

6. True or False: An increase in the use of automobiles can cause unusually cold weather due to higher carbon dioxide levels in the atmosphere.
7. True or False: A tornado creates a narrow path of destruction as it moves through an area, including forests.
8. True or False: Developing multiple hypotheses is the most crucial step in drawing conclusions from scientific investigations.
9. True or False: An apple core will decay faster than a glass bottle, a metal can, or a plastic bottle.
10. True or False: Alternative fuels are available at every gas station, making them a common choice for drivers.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. How can students effectively measure and communicate the size of oak tree leaves in a scientific context, and why is labeling an essential part of this process?
12. Discuss how you would effectively display and compare the percentages of minerals found in granite samples from western Nevada. What format would you choose and why?

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